Daniel O'Brien *Senior Software Engineer*

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Senior Software Engineer with 5+ years of experience architecting secure, high-scale cloud infrastructure and production AI systems. Expert in modernizing enterprise tech stacks using AWS (ECS Fargate, Secrets Manager), Terraform, and Docker, with a proven track record of optimizing pipelines to handle 10TB of daily data. Specialized in delivering business-critical full-stack solutions using Python, Go, and React while driving strict security compliance and operational efficiency. Outside of work, I am an avid outdoorsman who enjoys backpacking, rock climbing, and surfing.

WORK EXPERIENCE

SITE Technologies Nov 2024 – Present

Senior Software Engineer Chicago, Illinois

- Deployed machine learning models using FastAPI, Docker, and AWS ECS Fargate, utilizing Terraform for infrastructure-as-code and AWS Secrets Manager to secure sensitive credentials, ensuring strict compliance for processing 100GB of geospatial data daily
- Drove refactoring of existing data pipelines in Python using parallelization and algorithm optimization, reducing average processing times by 70% and accelerating delivery timelines
- Delivered full-stack mapping web applications using React and FastAPI to visualize geospatial data and perform real-time analysis, collaborating with product teams to improve user productivity by 500%
- Optimized computer vision machine learning models using PyTorch, improving accuracy by 25% and reducing inference times by 50% through strategic model architecture improvements

Aicadium Jun 2023 – Nov 2024

Software Engineer San Diego, California

- Led development of a full-stack web application using React, FastAPI, and Go microservices for image and video management, enabling customers to apply custom computer vision models for classification, object detection, and segmentation tasks, improving user productivity by 150%

- Built a high-throughput media catalog using DynamoDB to index metadata for 100,000+ daily assets (10TB), leveraging Redis to orchestrate asynchronous video processing pipelines and manage distributed state across microservices
- Architected a computer vision model training platform using FastAPI and PyTorch, deployed on Kubernetes for efficient scaling, reducing client onboarding time by 93% (from 2 weeks to 1 day) by automating model fine-tuning processes
- Championed secure cloud infrastructure design on AWS, implementing Application Load Balancers (ALB) for traffic distribution and AWS Secrets Manager for credential rotation; optimized multi-tenant isolation strategies to improve system scalability and reliability by 60%

SeekOps Jun 2022 – Jun 2023
Software Engineer Austin, Texas

- Pioneered development of an interactive 3D map application using React and WebSockets to stream real-time methane emissions data from drone flights, reducing data rendering latency by 80% and enabling live visualization for field operators
- Redesigned REST APIs to gRPC endpoints in Go for geospatial data ingestion, reducing average latency by 75% (from 200ms to 50ms) and decreasing error rates from 4.9% to 0.1%
- Integrated ClickHouse to perform high-speed OLAP analytics on massive streams of incoming drone IoT sensor data, optimizing Python ingestion pipelines to query billions of data points with sub-second latency
- Orchestrated deployment strategy for Python and Go microservices using Docker and Kubernetes on AWS, configuring ingress controllers and IAM roles to ensure least-privilege security access, improving deployment speed by 50%

Benson Hill Nov 2021 – Jun 2022
Data Engineer St. Louis, Missouri

LightBox Mar 2021 – Nov 2021
Data Engineer Irvine, California

Far Western Oct 2019 – Mar 2021
Geospatial Engineer Davis, California

SKILLS

Cloud Infrastructure — AWS (ECS Fargate, ALB, Secrets Manager, S3, Lambda, EC2, IAM), Terraform, Docker, Kubernetes, Google Cloud Platform, Azure

Backend Systems — Go, Python (FastAPI, Django), gRPC, WebSockets, RESTful APIs, Microservices, Distributed Systems

Databases — DynamoDB, ClickHouse, Redis, PostgreSQL, MySQL, MongoDB, Kafka, Neo4j, SQLite

Frontend Development — React, NextJS, TypeScript, JavaScript, Tailwind CSS, HTML/CSS

Machine Learning — PyTorch, TensorFlow, Scikit-Learn, Pandas, NumPy, Model Deployment

DevOps & Tools — CI/CD (GitHub Actions), Git, Agile/Scrum, Test-Driven Development (TDD), System Design

EDUCATION

University of California, Davis Sep 2016 – Jun 2019

Bachelor of Science, Environmental Science Davis, California

- Specialization in Geospatial Information Systems